JEE Mathematics: Quick Review with Grading and Practice Map - Sets & Relation (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (69 questions; a question that appears in two of these is counted once): High = 6 or more questions, Medium = 1 to 5 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Set and its Representation

Question 01

What is a set? What is its cardinal number?

Model Answer

A set is a well-defined collection of distinct, distinguishable objects (elements). $a \in A$: a is an element of A.

Cardinal number $n(A)$ (or $|A|$) = number of elements of A.

Ex: $A = \{x_1, x_2, x_3\} \Rightarrow n(A) = 3$.

Grade - Low

Practice map, main concept: Sheet: J-I 1

Also used in: none

Question 02

How is a set written in roster form and in set-builder form?

Model Answer

Roster: list the elements inside braces, e.g. $\{2, 4, 6\}$ (order and repetition do not matter).

Set-builder: $\{x : P(x)\}$ = set of all x for which the property $P(x)$ holds.

Ex: $\{x \in \mathbb{N} : x + 3y = 12 \text{ for some } y \in \mathbb{N}\} = \{9, 6, 3\}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 03

A set is given by an equation or inequality. How do you list or count its elements?

Model Answer

Solve the condition first, then keep only the values in the stated universe ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{R}$, $\mathbb{N} \times \mathbb{N}$...). Check the domain (square roots, $x \geq 0$, denominators).

For integer points in a region, fix one coordinate and count the other.

Ex: $\{x \in \mathbb{Z} : (x + 2)(x^2 - 5x + 6) = 0\} = \{-2, 2, 3\}$.

Grade - High

Practice map, main concept: Sheet: J-M 11, 17, 22, 33-34 | Question Bank: 7-8, 30

Also used in: Sheet: J-I 2, 7, 10, 12; J-M 23 | Question Bank: 1, 14

Types of Sets

Question 04

Define finite, infinite, singleton and null sets.

Model Answer

Finite: $n(A)$ is a natural number or 0. Infinite: the elements cannot be counted to an end, e.g. $\{x \in \mathbb{R} : 0 < x < 0.0001\}$.

Singleton: $n(A) = 1$. Null (empty, void) set $\phi = \{\}$: $n(\phi) = 0$.

Ex: $\{\phi\}$ and $\{0\}$ are singletons, not empty; $\{x \in \mathbb{R} : x^2 + x + 1 = 0\} = \phi$.

Grade - Low

Practice map, main concept: Sheet: J-I 2, 4-5, 10

Also used in: none

Question 05

What is the difference between equal sets and equivalent sets?

Model Answer

Equal: $A = B$ if they have exactly the same elements ($A \subseteq B$ and $B \subseteq A$). Equivalent: $n(A) = n(B)$ (finite sets).

Ex: $\{1, 2, 3\}$ and $\{a, b, c\}$ are equivalent, not equal.

Grade - Low

Practice map, main concept: none

Also used in: none

Subsets and Power Set

Question 06

What is a subset and a proper subset? How many subsets does a set with n elements have?

Model Answer

$A \subseteq B$ if $x \in A \Rightarrow x \in B$. Proper subset $A \subset B$: $A \subseteq B$ and $A \neq B$. ϕ and A are subsets of every A.

$$\text{Subsets} = {}^nC_0 + {}^nC_1 + \cdots + {}^nC_n = 2^n, \qquad \text{proper (or non-empty)} = 2^n - 1$$

Ex: $2^m - 2^n = 56 \Rightarrow 2^n(2^{m-n} - 1) = 8 \cdot 7 \Rightarrow m = 6,\ n = 3$.

Grade - Medium

Practice map, main concept: Sheet: J-I 3, 11 | Question Bank: 1, 21

Also used in: Sheet: J-M 22

Question 07

Define superset.

Model Answer

If $A \subseteq B$, then B is a superset of A, written $B \supseteq A$ (proper: $B \supset A$).

Ex: $\{a, e, i, o, u\} \supset \{i, o\}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 08
What is the power set of a set?

Model Answer
$P(A) = \{X : X \subseteq A\}$, the set of all subsets of A.

$$n(A) = n \Rightarrow n(P(A)) = 2^n, \qquad n(P(P(A))) = 2^{2^n}$$

Ex: $P(\phi) = \{\phi\}$, $P(P(\phi)) = \{\phi, \{\phi\}\}$, so $n(P(P(P(\phi)))) = 4$.

Grade - Low
Practice map, main concept: Sheet: J-I 6
Also used in: none

Question 09
How do you count subsets (or pairs of subsets) that satisfy a condition?

Model Answer
Decide each element independently: subsets containing k fixed elements $= 2^{n-k}$; meeting a set B: $2^n - 2^{n-n(A \cap B)}$ (use the complement).

Ordered pairs (Y, Z) of subsets with $Y \cap Z = \phi$: each element has 3 choices $\Rightarrow 3^n$.

Ex: $X = \{1, 2, 3, 4, 5\}$: number of (Y, Z) with $Y \cap Z = \phi$ is $3^5 = 243$.

Grade - Medium
Practice map, main concept: Sheet: J-M 9, 31-32 | Question Bank: 25
Also used in: Sheet: J-M 10

Operations on Sets and Venn Diagram

Question 10
What are the universal set and the complement of a set?

Model Answer
Universal set U: contains all sets under consideration. Complement $A' = A^c = \{x \in U : x \notin A\} = U - A$.

$$(A')' = A, \quad A \cup A' = U, \quad A \cap A' = \phi, \quad \phi' = U$$

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-I 17, 22; J-M 22

Question 11
Define union, intersection, difference and symmetric difference of two sets.

Model Answer
$A \cup B = \{x : x \in A \text{ or } x \in B\}$, $A \cap B = \{x : x \in A \text{ and } x \in B\}$.

$$A - B = \{x : x \in A,\ x \notin B\} = A \cap B', \qquad A \,\Delta\, B = (A - B) \cup (B - A)$$

Ex: $A = \{2, 3, 4, 5, 6, 7\}$, $B = \{3, 5, 7, 9, 11, 13\}$: $A - B = \{2, 4, 6\}$, $B - A = \{9, 11, 13\}$.

Grade - High
Practice map, main concept: Sheet: J-I 7-8, 12, 22; J-M 23
Also used in: Sheet: J-M 12, 19, 26-27, 33-34 | Question Bank: 8, 35

Question 12
When are two sets disjoint?

Model Answer
A, B are disjoint if $A \cap B = \phi$; then $n(A \cup B) = n(A) + n(B)$. Otherwise they are overlapping.
Ex: $A - B$, $A \cap B$, $B - A$ are pairwise disjoint and their union is $A \cup B$.

Grade - Low
Practice map, main concept: Sheet: J-I 15-16
Also used in: none

Question 13
State the laws of algebra of sets.

Model Answer
Idempotent $A \cup A = A$, $A \cap A = A$; identity $A \cup \phi = A$, $A \cap U = A$; commutative and associative for $\cup, \cap$.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \qquad A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Also useful: $A \cup (A \cap B) = A$; if $A \cup B = A \cup C$ and $A \cap B = A \cap C$ then $B = C$.
Ex: $X \cap (Y \cup X)' = X \cap Y' \cap X' = \phi$.

Grade - Medium
Practice map, main concept: Sheet: J-I 17; J-M 5
Also used in: Sheet: J-I 19

Question 14
State De Morgan's laws.

Model Answer

$$(A \cup B)' = A' \cap B', \qquad (A \cap B)' = A' \cup B'$$

Also $A - (B \cup C) = (A - B) \cap (A - C)$ and $A - (B \cap C) = (A - B) \cup (A - C)$.
Ex: $(A \cup B')' \cup (B \cap C) = (A' \cap B) \cup (B \cap C) = B \cap (A' \cup C)$.

Grade - Low
Practice map, main concept: Sheet: J-I 19
Also used in: Sheet: J-I 17-18

Question 15
How do you read a statement in a Venn diagram?

Model Answer
U is a rectangle, sets are circles; three sets give 8 regions (only A, only $A \cap B$, ..., none). Shade exactly the regions the statement describes.
Ex: "some play two games, none play all three": the regions for exactly two are non-empty and $A \cap B \cap C = \phi$.

Grade - Medium
Practice map, main concept: Sheet: J-M 15
Also used in: none

Number of Elements in Sets

Question 16

Write the formulas for the number of elements in the union, difference and complement of two sets.

Model Answer

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A - B) = n(A) - n(A \cap B), \qquad n(A \,\Delta\, B) = n(A) + n(B) - 2n(A \cap B)$$

$$n(A' \cap B') = n(U) - n(A \cup B), \qquad n(A' \cup B') = n(U) - n(A \cap B)$$

Ex: 100 students, 55 pass Maths, 67 pass Physics, none fail: both $= 22$, Physics only $= 67 - 22 = 45$.

Grade - Medium

Practice map, main concept: Sheet: J-I 18, 21

Also used in: Sheet: J-I 15-16 | Question Bank: 17

Question 17

Write the formula for $n(A \cup B \cup C)$. How do you find "exactly one" and "exactly two"?

Model Answer

$$n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$$

Exactly two $= \sum n(A \cap B) - 3n(A \cap B \cap C)$; exactly one $= \sum n(A) - 2\sum n(A \cap B) + 3n(A \cap B \cap C)$.

Ex: medals $48 + 25 + 18 = 91$ to 60 men, 5 got all three: $e_1 + e_2 = 55,\ e_1 + 2e_2 + 15 = 91 \Rightarrow e_2 = 21$.

Grade - Medium

Practice map, main concept: Sheet: J-I 14 | Question Bank: 28, 31

Also used in: none

Question 18

How do you find the greatest and least possible values of $n(A \cap B)$ or $n(A \cup B)$?

Model Answer

$$\max(0,\ n(A) + n(B) - n(U)) \le n(A \cap B) \le \min(n(A), n(B))$$

$$\max(n(A), n(B)) \le n(A \cup B) \le \min(n(A) + n(B),\ n(U))$$

Ex: 63% read A, 76% read B, x% both: $39 \le x \le 63$.

Grade - Medium

Practice map, main concept: Sheet: J-I 13, 20; J-M 20

Also used in: Question Bank: 28, 31

Question 19

How do you count (or add) the multiples of given numbers in a range?

Model Answer

Multiples of k in $1..N$: $\left[\frac{N}{k}\right]$. Multiples of a and b = multiples of $\text{lcm}(a, b)$, e.g. $N_6 \cap N_8 = N_{24}$. Use inclusion-exclusion for "or", complement for "neither".

Sets like $\{9k + 2\}$ inside a range are A.P.s: count with T_n, add with S_n.

Ex: in $1..140$, numbers divisible by none of $2, 3, 5$: $140 - 102 = 38$.

Grade - Medium

Practice map, main concept: Sheet: J-I 9; J-M 12, 19, 26 | Question Bank: 17

Also used in: none

Cartesian Product of two sets

Question 20

How is the cartesian product of sets A and B represented?

Model Answer

$A \times B$ (read "A cross B"); its elements are ordered pairs (a, b), listed in roster form or drawn as points/arrows.

Ex: $A = \{2, 3\}, B = \{1, 2\}$: $A \times B = \{(2, 1), (2, 2), (3, 1), (3, 2)\}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 21

Define cartesian product of sets A and B.

Model Answer

$$A \times B = \{(a, b) : a \in A,\ b \in B\}$$

$(a, b) = (c, d) \iff a = c,\ b = d$; in general $A \times B \neq B \times A$ (equal only if $A = B$ or one set is empty); $A \times \phi = \phi$.

Grade - Low

Practice map, main concept: Sheet: J-II 13

Also used in: none

Question 22

Find the number of elements in $A \times B$.

Model Answer

$$n(A \times B) = n(A) \cdot n(B), \qquad n(P(A \times B)) = 2^{mn}$$

Ex: $n(A) = 2,\ n(B) = 4$: subsets of $A \times B$ with at least 3 elements $= 2^8 - 1 - 8 - 28 = 219$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 10

Question 23

Which identities hold for the cartesian product?

Model Answer

$$A \times (B \cup C) = (A \times B) \cup (A \times C), \qquad A \times (B \cap C) = (A \times B) \cap (A \times C)$$
$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D), \qquad A \times (B - C) = (A \times B) - (A \times C)$$

Hence $A \times B$ and $B \times A$ have $n(A \cap B)^2$ elements in common. Ex: $A = \{a, b, c, d\}, B = \{b, c, d, e\}$: $3^2 = 9$.

Grade - Medium

Practice map, main concept: Sheet: J-I 23-25; J-M 10

Also used in: none

Relation

Question 24
Define relation.

Model Answer
A relation R from A to B is any subset of $A \times B$; $(a, b) \in R$ is written aRb. A relation on A is a subset of $A \times A$.
Ex: $\{(1,1),(2,1),(3,3)\} \subseteq X \times Y$ for $X = \{1,2,3\}, Y = \{1,3,5\}$ is a relation from X to Y.

Grade - Low
Practice map, main concept: Sheet: J-II 5
Also used in: none

Question 25
How can a relation from A to B be represented?

Model Answer
Roster form (list of ordered pairs), set-builder form $\{(x, y) : \text{rule}\}$, or an arrow diagram.
Ex: $R = \{(x, y) \in \mathbb{N}^2 : x + 3y = 12\} = \{(9,1),(6,2),(3,3)\}$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 5, 8-9, 17

Question 26
What do you mean by pre-image and image of an element in a relation?

Model Answer
If $(a, b) \in R$, then b is an image of a and a is a pre-image of b. One element may have many images (unlike a function).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 27
What is the domain of a relation?

Model Answer

$$\text{Dom}(R) = \{a : (a, b) \in R \text{ for some } b\}$$

= set of all first entries (pre-images). Ex: $R = \{(9,1),(6,2),(3,3)\}$: Dom $= \{3,6,9\}$.

Grade - Low
Practice map, main concept: Sheet: J-II 8
Also used in: none

Question 28
What is the co-domain of a relation?

Model Answer
For $R : A \to B$ the co-domain is the whole set B, whether or not each element is related.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 29
What is the range of a relation?

Model Answer

$$\text{Range}(R) = \{b : (a, b) \in R \text{ for some } a\} \subseteq \text{co-domain}$$

Ex: $A = \{1, 2, 3\}, B = \{1, 4, 6, 9\}$, $xRy \iff x > y$: $R = \{(2, 1), (3, 1)\}$, range $= \{1\}$.

Grade - **Medium**
Practice map, main concept: Sheet: J-II 17 | Question Bank: 14
Also used in: Sheet: J-II 8

Question 30
How many relations are there from A to B, and on A?

Model Answer

$$n(A) = m,\ n(B) = n: \quad \#\{R : A \to B\} = 2^{mn}, \qquad \#\{R \text{ on } A\} = 2^{m^2}$$

Ex: on $\{1, 2, 3\}$ there are $2^9 = 512$ relations.

Grade - **Low**
Practice map, main concept: Sheet: J-I 28
Also used in: none

Question 31
A relation is given by a rule. How do you count its elements?

Model Answer
Fix the first entry (or the value of an expression), count the allowed second entries, then add. For a relation on $A \times B$, the pairs are $((a_1, b_1), (a_2, b_2))$: count the choices of each of the 4 entries.
Ex: on $\{1, \ldots, 5\}$, $4x \leq 5y$: for $x = 1..5$ the counts are $5, 4, 3, 2, 2$, so $n(R) = 16$.

Grade - **High**
Practice map, main concept: Sheet: J-M 27, 30 | Question Bank: 2, 6, 29, 35
Also used in: Sheet: J-M 16 | Question Bank: 3-5, 16, 18, 34

Types of a relation

Question 32
What are the void relation and the universal relation? Which properties do they have?

Model Answer
Void $R = \phi$: symmetric and transitive, not reflexive (for $A \neq \phi$). Universal $R = A \times A$: reflexive, symmetric, transitive.

Every singleton relation $\{(a, b)\}$ is transitive.

Grade - **Low**
Practice map, main concept: Sheet: J-II 2
Also used in: none

Question 33
Define reflexive relation.

Model Answer

$$R \text{ on } A \text{ is reflexive} \iff (a,a) \in R \ \ \forall a \in A$$

Ex: on $\{1,2,3\}$, $\{(1,1),(2,2)\}$ is not reflexive ($(3,3)$ missing).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 34
How to check whether a given relation is reflexive?

Model Answer
Listed relation: all n pairs (a,a) must be present. Rule: put $y = x$ in the rule; it must hold for every x in the set.
Ex: $1 + ab > 0$: $1 + a^2 > 0$ always, so reflexive. a divides b on $\mathbb{Z}$: fails at $a = 0$.

Grade - High
Practice map, main concept: none
Also used in: Sheet: J-I 26-27, 29-35, 37-39; J-II 1, 3-4, 6-7, 10-12, 14-16; J-M 1-4, 6-8, 14, 18, 21, 24-25, 28-29 | Question Bank: 6, 11-13, 15, 19-20, 22, 24

Question 35
Define identity relation.

Model Answer

$$I_A = \{(a,a) : a \in A\}$$

Every element is related only to itself. Identity is reflexive, but a reflexive relation need not be the identity.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 36
Define symmetric relation.

Model Answer

$$R \text{ is symmetric} \iff (a,b) \in R \Rightarrow (b,a) \in R$$

Ex: perpendicularity of lines in a plane is symmetric (but not reflexive or transitive).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 37
How to check whether a given relation is symmetric?

Model Answer
Listed: every (a, b) must have its mirror (b, a). Rule: swap x, y; the rule must stay the same (or follow from the original).

Ex: $x^2 - 4xy + 3y^2 = 0 \iff x = y$ or $x = 3y$: $(3, 1) \in R$ but $(1, 3) \notin R$, so not symmetric.

Grade - High
Practice map, main concept: none
Also used in: Sheet: J-I 26-27, 29-35, 37-39; J-II 1-4, 6-7, 9-12, 14-16; J-M 1-4, 6-8, 14, 18, 21, 24-25, 28-29 | Question Bank: 6, 11-13, 15, 19-20, 22, 24, 34

Question 38
Define transitive relation.

Model Answer

$$R \text{ is transitive} \iff (a, b) \in R,\ (b, c) \in R \Rightarrow (a, c) \in R$$

Ex: "is a brother of" is transitive among the children of a family (not symmetric).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 39
How to check whether a given relation is transitive?

Model Answer
Listed: for every chain $(a, b), (b, c)$ check that (a, c) is present (pairs with $a = b$ or $b = c$ add nothing). Rule: try to prove it; otherwise find a counter-example with small numbers.

Ex: $1 + ab > 0$: $(1, \frac{1}{2}), (\frac{1}{2}, -1) \in R$ but $(1, -1) \notin R$, so not transitive.

Grade - High
Practice map, main concept: none
Also used in: Sheet: J-I 26-27, 29-34, 37-39; J-II 1-4, 6-7, 9-12, 14-16; J-M 1-4, 6-8, 14, 18, 21, 24-25, 28-29 | Question Bank: 6, 11-13, 15, 19-20, 22, 24, 26-27

Question 40
What do you mean by equivalence relation?

Model Answer
R is an equivalence relation if it is reflexive, symmetric and transitive. Test all three separately.

Typical equivalences: $a - b$ divisible by n; $a - b \in \mathbb{Z}$; $|x| = |y|$; parallel lines; congruent or similar triangles; similar matrices $B = P^{-1}AP$; $ad = bc$ on pairs.

Ex: $2a + 3b$ multiple of $5 \iff 5 \mid 5a - 3(a - b) \iff 5 \mid a - b$: an equivalence relation.

Grade - High
Practice map, main concept: Sheet: J-I 26-27, 29-34, 37-39; J-II 1, 3-4, 6-7, 9-12, 14-16; J-M 1-4, 6-8, 14, 18, 21, 24-25, 28-29 | Question Bank: 11-13, 15, 19-20, 22, 24
Also used in: Sheet: J-I 36 | Question Bank: 10, 23, 33

Question 41
Define anti-symmetric relation.

Model Answer

$$R \text{ is anti-symmetric } \iff (a,b) \in R \text{ and } (b,a) \in R \Rightarrow a = b$$

Ex: $\subseteq$, $\leq$, "divides" on $\mathbb{N}$ (not on $\mathbb{Z}$: $2 \mid -2,\ -2 \mid 2$).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 42
What is the equivalence class of an element in a relation?

Model Answer

$$[a] = \{x \in A : xRa\}$$

Equivalence classes are disjoint and their union is A (a partition); conversely every partition $A = A_1 \cup \cdots \cup A_k$ gives "same block" as an equivalence relation.

Ex: "same distance from O": class of $(1,-1)$ is the circle $x^2 + y^2 = 2$.

Grade - Medium
Practice map, main concept: Sheet: J-M 13, 16
Also used in: Sheet: J-M 25 | Question Bank: 11

Question 43
How many reflexive, symmetric and equivalence relations can be defined on a set of n elements?

Model Answer

$$\text{reflexive} : 2^{n(n-1)}, \quad \text{symmetric} : 2^{\frac{n(n+1)}{2}}, \quad \text{reflexive and symmetric} : 2^{\frac{n(n-1)}{2}}$$

Anti-symmetric: $2^n \cdot 3^{\frac{n(n-1)}{2}}$. Equivalence relations = Bell number B_n ($B_3 = 5,\ B_4 = 15$).

Ex: symmetric but not reflexive on 4 elements: $2^{10} - 2^6 = 960$.

Grade - Medium
Practice map, main concept: Question Bank: 9, 26-27, 32
Also used in: none

Question 44
What is the least number of pairs to add so that a relation becomes reflexive / symmetric / transitive / an equivalence?

Model Answer

Reflexive: add the missing (a,a). Symmetric: add the missing mirrors (b,a). Transitive: keep adding (a,c) for every chain until none is missing.

Equivalence: join the related elements into blocks; the smallest equivalence has $\sum s_i^2$ pairs (s_i = block sizes); subtract the pairs already present.

Ex: $R = \{(1,2),(2,3)\}$ on $\{1,2,3\}$: one block of 3, $9 - 2 = 7$ pairs to add.

Grade - High
Practice map, main concept: Sheet: J-I 35-36 | Question Bank: 3-5, 10, 16, 18, 23, 33-34
Also used in: none

Inverse of a relation

Question 45

What is inverse of a relation?

Model Answer

$$R^{-1} = \{(b, a) : (a, b) \in R\}, \qquad \text{Dom}(R^{-1}) = \text{Range}(R), \quad \text{Range}(R^{-1}) = \text{Dom}(R)$$

Ex: $R = \{(2, 4), (4, 3), (6, 2), (8, 1)\} \Rightarrow R^{-1} = \{(4, 2), (3, 4), (2, 6), (1, 8)\}$. R is symmetric $\iff R = R^{-1}$.

Grade - Low

Practice map, main concept: none

Also used in: none